

ECG Classification

The objective of this project is to classify ECG signals sourced from Kaggle's dataset. Three different approaches were employed, consisting of two traditional machine learning methods and one deep learning method.

1. PCA + SVM:

- **Feature Extraction:** Employed PCA for dimension reduction and extracting relevant features from the ECG signals.
- **Classification:** Implemented Support Vector Machines (SVM) for classification.

2. PCA + MLP:

- **Feature Extraction:** Employed PCA for dimension reduction and extracting relevant features from the ECG signals.
- **Classification:** Utilized Multilayer Perceptron (MLP) for classification.

3. Convolutional Neural Network (CNN):

- Implemented a Convolutional Neural Network, a deep learning approach renowned for its effectiveness in capturing intricate patterns within data.

Dataset Information

This dataset contains the ECG readings of patients. Each row corresponds to a single complete ECG of a patient. Every single ECG is composed of 140 data points.

Columns:

- 1) Columns 0-139 contain the ECG data point for a particular patient. These are floating point numbers.
- 2) The label which shows whether the ECG is normal or abnormal. It is a categorical variable with value either 0 or 1.

Here I plot one sample of each group:

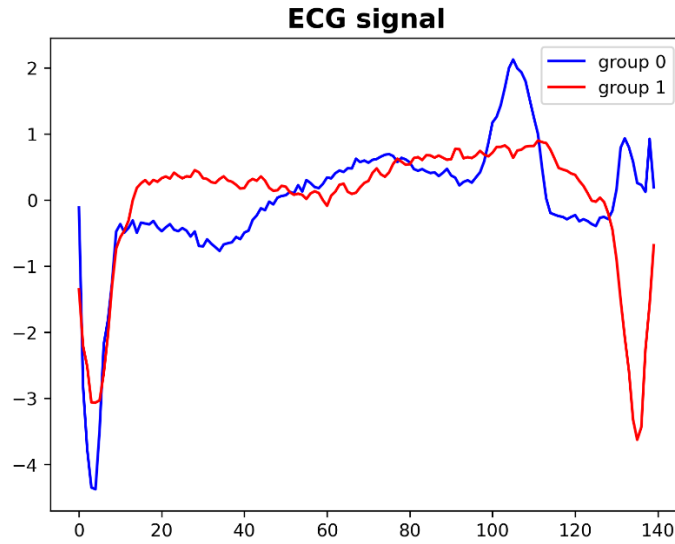


Figure1: ECG signal in two groups

ECG classification using traditional machine learning methods

Feature extraction

Since dimension of the ECG signal is relatively large, it is better to extract features from the original signal rather than using the signal directly as input for methods like SVM or MLP. This helps to prevent overfitting.

As depicted in the above figure, temporal patterns of ECG signals differ in two groups. Therefore, we should consider this in our feature extraction process. Statistical features like mean, variance, kurtosis, or skewness may not effectively capture the temporal information of signals in two groups. Consequently, less discriminative features would be extracted.

To illustrate this point, I extracted the mean and variance of the ECG signal and plotted them below.

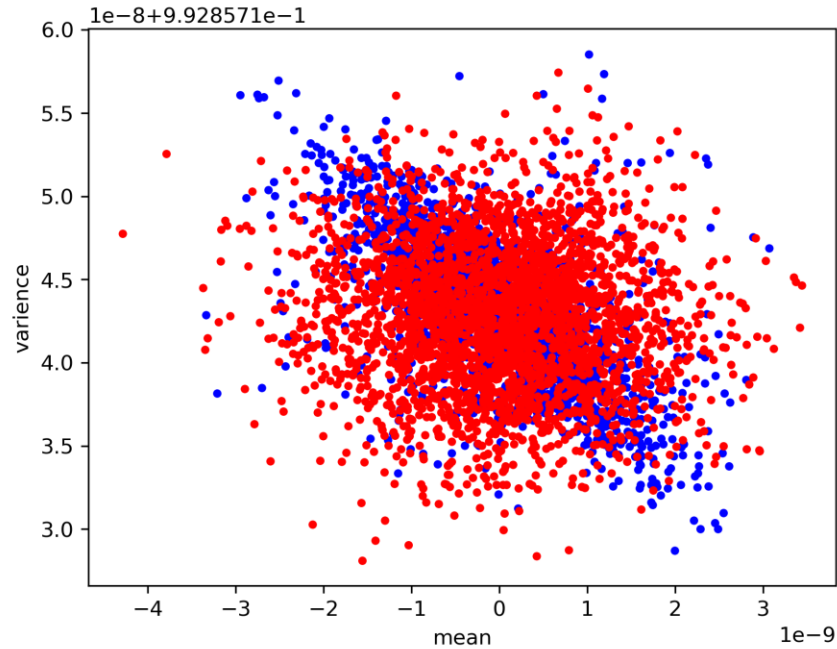


Figure 2: 2D plot of groups based on statistical features

As shown in the figure 2, the two groups are not clearly separable based on these statistical features.

PCA for feature extraction

Two approaches can be efficient for feature extraction and dimensionality reduction: down-sampling and Principal Component Analysis (PCA). Since down-sampling may not be efficient in reducing more dimensions, and, on the other side, PCA preserves the inherent and most informative aspects of the ECG signal, I employed PCA in this project for feature extraction and dimension reduction.

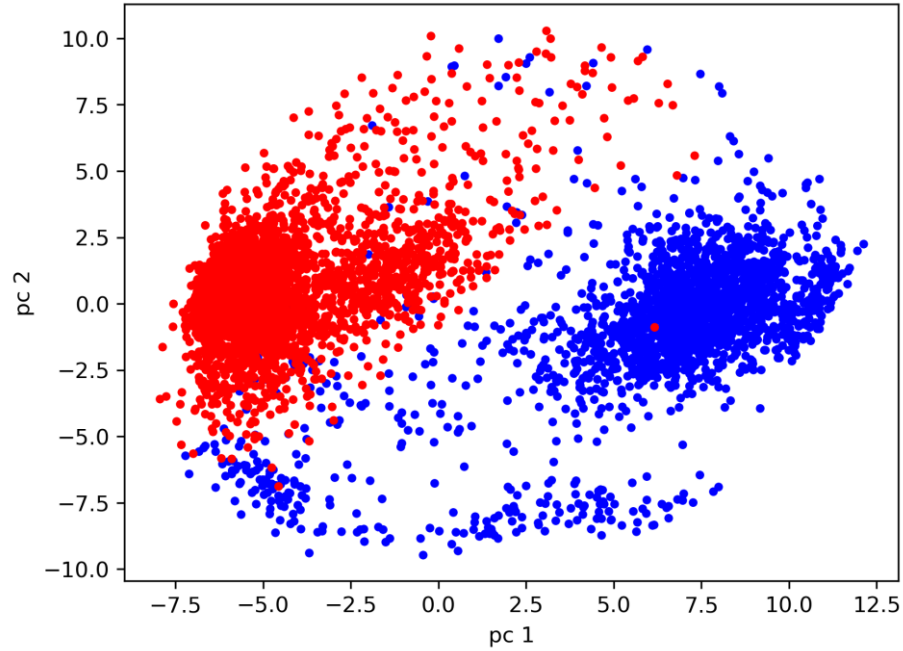


Figure 3: 2D plot of groups based on PCA features

As shown in Figure 3, the two groups are clearly separable based on these features. We reduced the dimension to 2 dimensions to illustrate the effectiveness of PCA. In practice, we can consider more principal components in constructing features, such as 3 or more dimensions.

Classification

Considering the generality and robustness demonstrated by SVM and MLP in classification, I utilized these classifiers for the classification task, and the results are presented in Table 1.

<i>Model</i>	<i>Accuracy</i>	<i>Sensitivity</i>	<i>Specificity</i>	<i>Confusion matrix</i>
SVM	97.867%	95.411%	99.654%	[[603 29] [3 865]]
MLP	98.333%	96.835%	99.424%	[[612 20] [5 863]]

ECG classification using Convolutional Neural Network (CNN)

As I mentioned, temporal patterns of ECG signals differs between the two groups and includes meaningful information. On the other hand, CNN's convolutional layers are designed to adeptly capture spatial and temporal information, makes it an ideal choice for this task. The intricate

architecture of CNN allows it to effectively learn and discern meaningful features within the temporal patterns, contributing to its efficacy in ECG signal classification.

I used the following architecture for my CNN.

Input	Convolution	Pooling	Full connect1	Full connect2	Full connect3
ECG signal (1x140)	Conv1d kernel_size=5 padding: 2 in_channels=1 out_channels=10	MaxPool1d kernel_size=4 stride=4	in_channels=20x35 out_channels=120	in_channels=120 out_channels=80	in_channels=80 out_channels=1

The classification results of CNN are presented in Table 2.

Model	Accuracy	Sensitivity	Specificity	Confusion matrix
CNN	99.533%	99.367%	99.654%	[[628 4] [3 865]]

As expected, Convolutional Neural Network (CNN) demonstrated superior performance compared to other methods employed in the project. The network's ability to effectively capture and analyze the intricate temporal patterns of ECG signals contributed to its outstanding classification performance. This highlights the suitability of CNN for tasks involving ECG signal classification, showcasing its potential as a robust and effective method in this domain.